

Questions

MathonGo

Q1 - 24 June - Shift 2

The number of solutions of the equation

$$\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x, \quad x \in [-3\pi,$$

$3\pi]$ is :

(A) 8

(B) 5

(C) 6

(D) 7

Space for your notes:

Q2 - 25 June - Shift 2

The value of $2\sin(12^\circ) - \sin(72^\circ)$ is :

(A) $\frac{\sqrt{5}(1-\sqrt{3})}{4}$

(B) $\frac{1-\sqrt{5}}{8}$

(C) $\frac{\sqrt{3}(1-\sqrt{5})}{2}$

(D) $\frac{\sqrt{3}(1-\sqrt{5})}{4}$

Space for your notes:

Q3 - 26 June - Shift 1

If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha -$

$\frac{1}{16}\sin(10^\circ)$, then $16 + \alpha^{-1}$ is equal to _____.

Space for your notes:

Q4 - 26 June - Shift 2

$16\sin(20^\circ)\sin(40^\circ)\sin(80^\circ)$ is equal to :

(A) $\sqrt{3}$ (B) $2\sqrt{3}$

(C) 3

(D) $4\sqrt{3}$

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Q5 - 27 June - Shift 1

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The value of $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is equal to :

- (A) -1 (B) $-\frac{1}{2}$
(C) $-\frac{1}{3}$ (D) $-\frac{1}{4}$

Space for your notes:

Q6 - 27 June - Shift 2

$\alpha = \sin 36^\circ$ is a root of which of the following equation

- (A) $10x^4 - 10x^2 - 5 = 0$ (B) $16x^4 + 20x^2 - 5 = 0$
(C) $16x^4 - 20x^2 + 5 = 0$ (D) $16x^4 - 10x^2 + 5 = 0$

Space for your notes:

Q7 - 28 June - Shift 2

If $\cot \alpha = 1$ and $\sec \beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$

and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and

the quadrant in which $\alpha + \beta$ lies, respectively are

- (A) $-\frac{1}{7}$ and IVth quadrant
(B) 7 and Ist quadrant
(C) -7 and IVth quadrant
(D) $\frac{1}{7}$ and Ist quadrant

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Answer Key

Q1 (D)

Q2 (D)

Q3 (80)

Q4 (B)

Q5 (B)

Q6 (C)

Q7 (A)

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Q1 (D)

$$\cos\left(\frac{\pi}{3} + x\right) \cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4} \cos^2 2x$$

$$x \in [-3\pi, 3\pi]$$

$$4\left(\cos^2\left(\frac{\pi}{3}\right) - \sin^2 x\right) = \cos^2 2x$$

$$4\left(\frac{1}{4} - \sin^2 x\right) = \cos^2 2x$$

$$1 - 4 \sin^2 x = \cos^2 2x$$

$$1 - 2(1 - \cos 2x) = \cos^2 2x$$

$$\text{let } \cos 2x = t$$

$$-1 + 2 \cos 2x = \cos^2 2x$$

$$t^2 - 2t + 1 = 0$$

$$(t-1)^2 = 0$$

$$\boxed{t=1} \quad \boxed{\cos 2x=1}$$

$$2x = 2n\pi$$

$$\boxed{x = n\pi}$$

$$n = -3, -2, -1, 0, 1, 2, 3$$

(D) option is correct.

Q2 (D)

$$\begin{aligned}
 & \sin 12^\circ + \sin 12^\circ - \sin 72^\circ \\
 &= \sin 12^\circ - 2 \cos 42^\circ \sin 30^\circ \\
 &= \sin 12^\circ - \sin 48^\circ \\
 &= -2 \cos 30^\circ \sin 18^\circ \\
 &= -2 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{5}-1}{4} \\
 &= \frac{\sqrt{3}}{4} (1 - \sqrt{5})
 \end{aligned}$$

Q3 (80)

$$\begin{aligned}
 & \sin 10^\circ \left(\frac{1}{2} \cdot 2 \sin 20^\circ \sin 40^\circ \right) \cdot \sin 10^\circ \sin(60^\circ - 10^\circ) \sin(60^\circ + 10^\circ) \\
 &= \sin 10^\circ \cdot \frac{1}{2} (\cos 20^\circ - \cos 60^\circ) \cdot \frac{1}{4} \sin 30^\circ \\
 &= \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \sin 10^\circ \left(\cos 20^\circ - \frac{1}{2} \right) \\
 &= \frac{1}{32} (2 \sin 10^\circ \cos 20^\circ - \sin 10^\circ) \\
 &= \frac{1}{32} (\sin 30^\circ - \sin 10^\circ - \sin 10^\circ) \\
 &= \frac{1}{32} \left(\frac{1}{2} - 2 \sin 10^\circ \right) \\
 &= \frac{1}{64} (1 - 4 \sin 10^\circ) \\
 &= \frac{1}{64} - \frac{1}{16} \sin 10^\circ \\
 & \text{Hence } \alpha = \frac{1}{64}
 \end{aligned}$$

Q4 (B)

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Hints and Solutions

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$$\begin{aligned}
 &16 \sin 20^\circ \sin 40^\circ \sin 80^\circ \\
 &= 16 \sin 40^\circ \sin 20^\circ \sin 80^\circ \\
 &= 4(4 \sin (60 - 20) \sin (20) \sin (60 + 20)) \\
 &= 4 \times \sin (3 \times 20^\circ) \\
 &[\because \sin 3\theta = 4 \sin(60 - \theta) \times \sin \theta \times \sin (60 + \theta)] \\
 &= 4 \times \sin 60^\circ \\
 &= 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}
 \end{aligned}$$

Q5 (B)

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$$\begin{aligned} & \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \\ &= \frac{\sin \left(3 \times \frac{\pi}{7} \right)}{\sin \frac{\pi}{7}} \times \cos \left(\frac{\frac{2\pi}{7} + \frac{6\pi}{7}}{2} \right) \\ &= \frac{2 \sin \left(\frac{3\pi}{7} \right)}{2 \sin \frac{\pi}{7}} \times \cos \left(\frac{4\pi}{7} \right) \\ &= \frac{\sin \left(\frac{7\pi}{7} \right) + \sin \left(\frac{-\pi}{7} \right)}{2 \sin \frac{\pi}{7}} \\ &= \frac{-\sin \frac{\pi}{7}}{2 \sin \frac{\pi}{7}} \\ &= -\frac{1}{2} \end{aligned}$$

Q6 (C)

$$\cos 72^\circ = \frac{\sqrt{5}-1}{4}$$

$$\Rightarrow 1 - 2 \sin^2 36^\circ = \frac{\sqrt{5}-1}{4}$$

$$\Rightarrow 4 - 8\alpha^2 = \sqrt{5} - 1$$

$$\Rightarrow 5 - 8\alpha^2 = \sqrt{5}$$

$$\Rightarrow (5 - 8\alpha^2)^2 = 5$$

$$\Rightarrow 25 + 64\alpha^4 - 80\alpha^2 = 5$$

$$\Rightarrow 64\alpha^4 - 80\alpha^2 + 20 = 0$$

$$\Rightarrow 16\alpha^4 - 20\alpha^2 + 5 = 0$$

Q7 (A)

$$\cot \alpha = 1, \sec \beta = \frac{-5}{3}, \cos \beta = \frac{-3}{5}, \tan \beta = \frac{-4}{3}$$

$$\tan(\alpha + \beta) = \frac{1 - \frac{4}{3}}{1 + \frac{4}{3} \times 1} = \frac{-1}{7}$$